

SEARCHING

Q10)

Question:

DS	Session	Searching	Question Information	Level 1 Challenge 10
Question Description: An array A contains integers with the following constraints: 'A' contains elements in sorted order. Integer i occurs $i \cdot \text{floor}(\sqrt{i}) + \text{ceil}(i/2)$ times in the array. All elements are greater than or equal to 1. Fathima has given Q queries of type: L R: Find the number of distinct values present in subarray A[L...R]. Note: 1-based indexing is followed. Constraints: $1 \leq Q \leq 10^5$ $1 \leq L \leq R \leq 10^{13}$ Input Format: The first line contains an integer Q denoting the number of queries. Next Q lines contains two space-separated integers L, R, denoting the query. Output Format: Print the output in separate lines contains required number of distinct values. Example Sample Input: 2 1 3 1 6 Sample Output: 2				

PROGRAM:

```
1  #include <stdio.h>
2  #include <math.h>
3
4  double a[300001];
5
6  int main() {
7      int q, i, l, r, m, x, y;
8      double L, R;
9
10     a[0] = 0;
11     for (i = 1; i <= 300000; i++)
12         a[i] = a[i-1] + i * (int)sqrt(i) + (i+1)/2;
13
14     scanf("%d", &q);
15
16     while (q--) {
17         scanf("%lf %lf", &L, &R);
18
19         l = 1;
20         r = 300000;
21         while (l < r) {
22             m = (l + r) / 2;
23             if (a[m] >= L)
24                 r = m;
25             else
26                 l = m + 1;
27         }
28         x = l;
29
30         l = 1;
31         r = 300000;
32         while (l < r) {
33             m = (l + r) / 2;
34             if (a[m] >= R)
35                 r = m;
36             else
37                 l = m + 1;
38         }
39         y = l;
40
41         if (L > R)
42             printf("1\n");
43         else
44             printf("%d\n", y - x + 1);
45     }
46
47     return 0;
48 }
```

OUTPUT:

TEST CASE 1:

```
2
3 3
1
2 6
3
```

TEST CASE 2:

```
3
3 3
1
2 6
3
5 4
1
```